



Practical Organic Chemistry : RACE-01

Test of Functional Group

1. Compounds I and II can be distinguished by using which reagent ?

4-Hydroxy-4-methylpent-2-enoic acid	5-Hydroxypent-2-ynoic
(I)	(II)

(A) NaHCO_3

(B) $\text{Br}_2/\text{H}_2\text{O}$

(C) HCl/ZnCl_2 (anhydrous)

(D) $\text{Cu}_2\text{Cl}_2/\text{NH}_4\text{OH}$

CP0042

2. Acetaldehyde and benzaldehyde can be differentiated by:

(A) Fehling test

(B) Iodoform test

(C) Tollen's reagent

(D) both (A) and (B)

CP0182

3. Which of the following reagent is used for the separation of acetaldehyde from acetophenone?

(A) NH_2OH

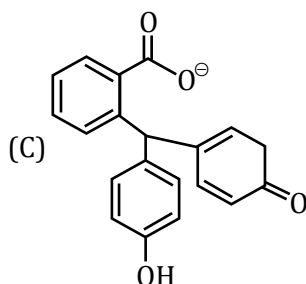
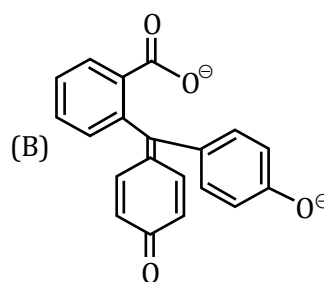
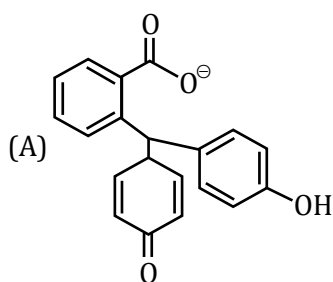
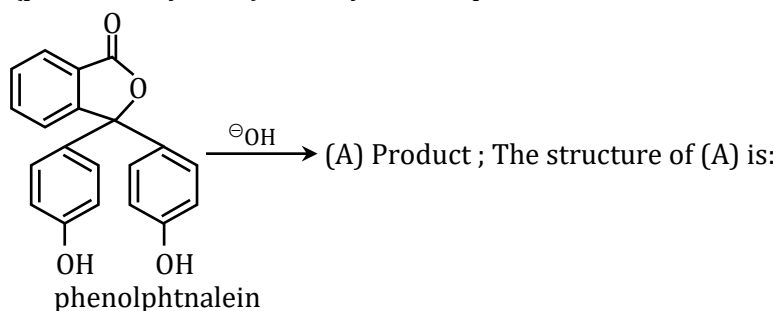
(B) NaOI

(C) Tollen's reagent

(D) $\text{C}_6\text{H}_5\text{NHNH}_2$

CP0184

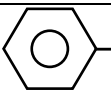
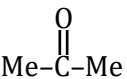
4. A dye, phenolphthalein is prepared by reacting phenol with phthalic anhydride in acidic medium. It give pink colour in alkaline medium due to extended conjugation in a new complex formed (phthalein-dye test) identify the complex A



(D) None

CP0185

5. Match the Column-I and Column-II

	Column-I (reagents)		Column-II
(a)	Product of reaction of propyne with 1% HgSO_4 and Dil. H_2SO_4	(p)	$\text{Me}-\text{CH}=\text{CH}-\text{Me}$
(b)	Br_2 water test given by	(q)	$\text{Me}-\text{C}\equiv\text{C}-\text{H}$
(c)	Addition of HBr can be with	(r)	
(d)	Tollen's reagent given white ppt. with	(s)	$\text{Me}-\text{CH}=\text{CH}_2$
		(t)	

(A) (a→t), (b→pqrs), (c→pqrs), (d→q)

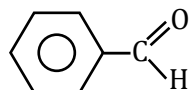
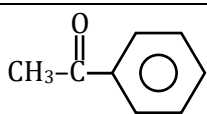
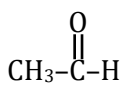
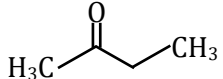
(B) (a→t), (b→pqrs), (c→rst), (d→p)

(C) (a→t), (b→qrs), (c→pqr), (d→r)

(D) (a→s), (b→pqrs), (c→st), (d→q)

CP0186

6. Match the Column-I and Column-II

	Column-I (Compound)		Column-II (tests)
(a)		(p)	2,4-DNP test
(b)		(q)	Yellow ppt with $\text{NaOH}+\text{I}_2$
(c)		(r)	Red ppt. with Fehling's solution
(d)		(s)	Silver mirror with Tollen's reagent

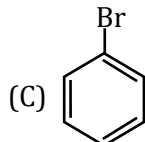
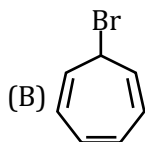
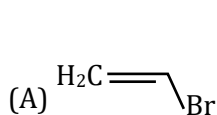
(A) (a→ps), (b→pq), (c→pqrs), (d→s)

(B) (a→ps), (b→pq), (c→pqrs), (d→pq)

(C) (a→ps), (b→pq), (c→qrs), (d→r)

(D) (a→rs), (b→rq), (c→p), (d→pq)

CP0187

7. Among the following which compound gives precipitate with AgNO_3 solution?

(D) None of these

CP0188

8. Phenol + Phthalic anhydride $\xrightarrow[\text{fusion}]{\text{Conc. H}_2\text{SO}_4}$ B. B gives which of the following colour in alkaline medium?

(A) Yellow

(B) Colourless

(C) Pink

(D) Violet

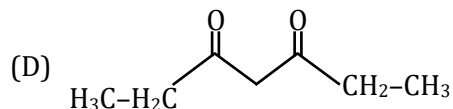
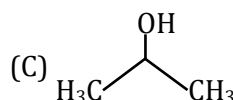
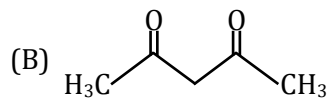
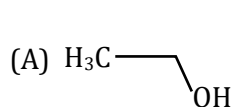
CP0189

9. Test to differentiate between (CH_3OH) and $(\text{Ph}-\text{OH})$ is/are :
(methanol) (Phenol)

- (A) Litmus test (B) FeCl_3
 (C) $\text{Br}_2 / \text{H}_2\text{O}$ (D) All of these

CP0190

10. Among the following, which will respond to iodoform test?



CP0191

11. Phenol and benzoic acid can be separated by

- (A) aq.KOH (B) NaOH solution
 (C) FeCl_3 solution (D) None of these

CP0214

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11
Answer	C	D	C	B	A	B	B	C	D	ABCD	BC